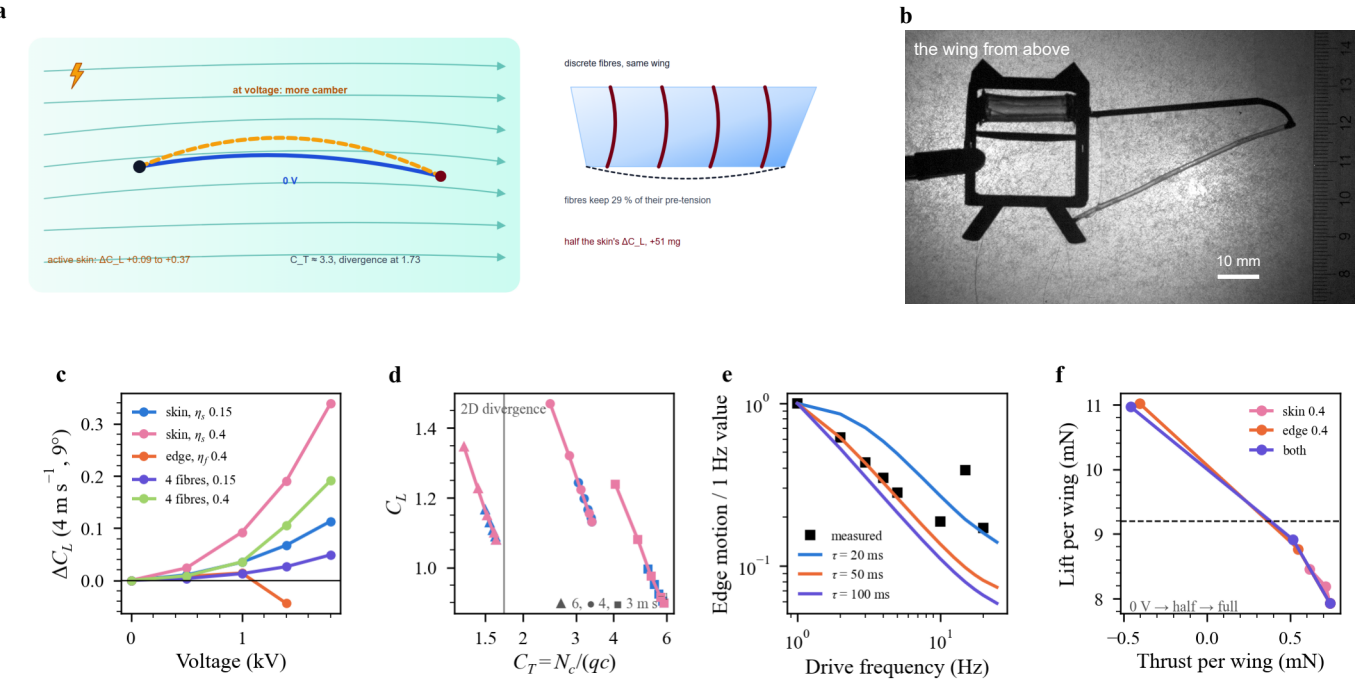

a
at voltage: more camber
0 V
active skin: ΔC_L +0.09 to +0.37
C_T ≈ 3.3, divergence at 1.73
discrete fibres, same wing
fibres keep 29 % of their pre-tension
half the skin's ΔC_L +51 mg
b
the wing from above
10 mm
c
skin, η_s 0.15
skin, η_s 0.4
edge, η_f 0.4
4 fibres, 0.15
4 fibres, 0.4
ΔC_L (4 m s^{-1}, 9°)
Voltage (kV)
d
2D divergence
▲ 6, ● 4, ■ 3 m s⁻¹
C_L
$C_T = N_c/(qc)$
e
measured
τ = 20 ms
τ = 50 ms
τ = 100 ms
Edge motion / 1 Hz value
Drive frequency (Hz)
f
skin 0.4
edge 0.4
both
0 V → half → full
Lift per wing (mN)
Thrust per wing (mN)